
Uncovering Conceptual Blindspots in Generative Image Models Using Sparse Autoencoders

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Abstract

Generative image models attain remarkable fidelity on large-scale training corpora, yet they systematically fail to render seemingly elementary concepts—human hands or objects appearing in groups of four, for instance—that are, by any reasonable account, well represented in their training data. Because such failures have to date been catalogued only anecdotally, it has remained unclear whether they are isolated idiosyncrasies or the surface manifestations of a broader, structural limitation. We take up this question by introducing a systematic, model-agnostic framework for identifying and characterizing “conceptual blindspots”: concepts that are prevalent in a reference distribution of natural image–caption pairs yet under- or over-represented in a model’s generations conditioned on the same captions. The framework uses sparse autoencoders (SAEs) to extract interpretable concept embeddings, reducing the comparison of a generative model against this reference distribution to a tractable, per-concept comparison of concept prevalence between paired real and generated images. To operationalize the comparison at scale, we train an archetypal SAE (RA-SAE) on DINOv2 features spanning 32,000 concepts—to our knowledge the largest such SAE reported to date—which affords fine-grained resolution of conceptual disparities. Applied to four widely used generative models (Stable Diffusion 1.5/2.1, PixArt, and Kandinsky), the method surfaces concretely suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated ones (e.g., wood background texture and palm trees), a representative subset of which we corroborate through targeted stress testing. Descending to the level of individual datapoints, we further isolate memorization artifacts—instances in which models reproduce highly specific visual templates encountered during training. Taken together, these results show that SAE-derived concept comparisons furnish a principled and practical diagnostic that quantifies a model’s conceptual fidelity relative to a reference image distribution, and that the blindspots it uncovers are systematic rather than anecdotal.

1 Introduction

Generative image models trained on large-scale datasets have attained unprecedented capabilities, underpinning applications both within the vision domain [1–9] and well beyond it [9–15]. This success notwithstanding, a growing body of qualitative [16–18] and quantitative [19, 20] evidence documents that these models nonetheless struggle to render conceptually elementary content, including human hands [21–24], objects appearing in groups of four [25], and negations or object relations [20, 26]. Characteristically, when prompted for such concepts, a model produces an output that is structurally adjacent to, yet distinct from, the intended target—rendering hands



Figure 1: The energy difference $\delta(k)$ quantifies the extent to which a model over- or under-generates a concept c_k relative to its prevalence in a reference distribution of natural images. Following the thresholds fixed throughout, concepts with $\delta(k) < 0.1$ are designated **suppressed conceptual blindspots** and those with $\delta(k) > 0.9$ **exaggerated conceptual blindspots**. The images shown, produced by four widely used generative image models, illustrate both regimes of conceptual blindspot and, for contrast, faithfully aligned concepts.

with six fingers, for example. We term these failure modes “conceptual blindspots”¹ they are counterintuitive precisely because the models have, in principle, been exposed to abundant demonstrations that faithfully depict the affected concepts. This tension motivates the central question of our work: are such failures idiosyncratic quirks confined to a handful of concepts, or the visible symptoms of a broader phenomenon in which, across many concepts, models are systematically disinclined—or, conversely, overly inclined—to generate the corresponding content?

Answering this question lies beyond the reach of existing approaches to evaluating generative image models [28, 29], each of which is limited along precisely the dimension our question demands. Realism-oriented measures such as FID [30] are inherently coarse-grained and therefore blind to concept-level, distributional failures. Diversity- and coverage-oriented statistics, exemplified by CLIPScore [31–33], afford only partial purchase, since they operate over whole images or prompts rather than at the granularity of individual features or concepts [34, 35] and thus cannot localize a blindspot to a specific concept. Qualitative protocols, finally—whether participant surveys [5, 17, 36–38] or open-ended exploration [39]—can surface individual failures, but supply no scalable, reproducible procedure that transfers across model families and supports systematic comparison among them.

This work. These limitations jointly delineate the methodology the problem requires: an automated, unsupervised procedure that elicits, directly from a reference image distribution, precisely those concepts whose odds of generation under a trained model diverge from their odds in that reference. We develop such a methodology and, across the four model families we evaluate, present evidence that conceptual blindspots are a systematic rather than a merely anecdotal phenomenon. Our contributions are threefold.

- **A formalization of conceptual blindspots.** We cast the identification of conceptual blindspots as a per-concept comparison between a model’s generations and a reference distribution of natural images, and we make this comparison precise through an energy-difference score with clear operational thresholds (Section 3). Grounding the evaluation in a reference image distribution, rather than in anecdotal or hand-curated concept sets, provides a principled and reproducible basis for characterizing model limitations at the level of individual concepts.
- **A scalable, unsupervised methodology built on sparse autoencoders.** We instantiate the formalization as a structured pipeline that uses sparse autoencoders (SAEs) to extract and compare concept distributions between natural and synthesized images (Section 4). Its core instrument is an archetypal SAE (RA-SAE) that we train and open-source on DINOv2 features spanning 32,000 concepts—to our knowledge the largest such RA-SAE reported to date—chosen for the seed-stability and data-faithfulness it confers on the recovered concept basis.
- **An exploratory tool and a cross-model empirical analysis.** We release an interactive web tool supporting both distribution- and datapoint-level analysis of blindspots (Sections 5.1–5.4), and we apply the method to Kandinsky, PixArt, and Stable Diffusion (SD) 1.5/2.1 (Section 5). The analysis pinpoints concrete suppressed blindspots (e.g., bird feeders, DVD discs, and whitespaces on documents) and exaggerated ones (e.g., wood background texture and palm trees), shown in Fig. 1 and distinguishes blindspots shared across models from those specific to individual architectures.

¹We borrow the term “blindspots” from the psychology literature [27], where it denotes situations in which an agent makes biased decisions despite exposure to observations that contradict the decision’s rationale.

78 The model, code, and web tool are available at a software repository.

79 2 Related Work

80 2.1 Explainability in Vision

81 Early work in explainable AI for computer vision concentrated on attributing predictions to influential
82 input regions [40–42]. Such attributions, however, convey little about the semantics of the learned
83 representation and can yield demonstrably incorrect explanations [43–45]. Concept-based inter-
84 pretability [46] arose to redress these shortcomings by identifying semantically meaningful directions
85 in a network’s representation, revealing not merely *where* a model attends but *which* concepts and
86 structures it employs [47–49].

87 A unifying insight of recent work is that the leading concept-based interpretability methods—
88 ACE [44], CRAFT [48], and sparse autoencoders (SAEs) [50–51]—instantiate a common dictionary-
89 learning objective under differing constraints [52]. Among these, SAEs have proven the most scalable.
90 The original formulation carries documented limitations—overly specific features [53], composition-
91 ality challenges [54], and limited intervention effects [55]—which have in turn motivated improved
92 variants, notably the archetypal SAE (RA-SAE) [56] that we adopt here.

93 Complementary interpretability approaches—prompt-based probing [57], attention-map and acti-
94 vation visualizations [58–59], and dataset-level statistics such as diversity or distribution-coverage
95 metrics [32–33]—each afford only partial answers to the question we pose. The decisive limitation is
96 one of granularity: they operate over full images or prompts rather than fine-grained features and
97 concepts [34–35], and so cannot localize which concept is at fault. Many further rely on subjective
98 interpretation and do not distinguish among distinct failure modes [60]. Existing methods and metrics
99 are consequently ill-suited to the systematic identification of feature- and concept-level weaknesses
100 in generative image models [28]—the gap our framework is designed to close.

101 2.2 Generative Image Models

102 Diffusion-based methods have become the dominant paradigm across generative vision modalities,
103 spanning image [4–61–64], video [65–72], and 3D [7–73–75] synthesis. In image generation
104 specifically, the lineage originates with denoising diffusion probabilistic models (DDPMs) [76] and
105 extends to their non-Markovian generalization, denoising diffusion implicit models (DDIMs) [77].

106 The Stable Diffusion (SD) [9] family brought DDPMs to broad accessibility across research and
107 open-source communities alike. The original SD spawned successive versions—SD 2 [78], SD 3 [79],
108 and SD XL [80]—alongside numerous architectural extensions that impose additional constraints on
109 the diffusion process (e.g., style [81–82], pose [83], and identity [84–85]) and admit alternative input
110 modalities such as image-to-image generation. A broader ecosystem of latent diffusion models has
111 since emerged atop SD, including Kandinsky [86], PixArt [87], DeepFloyd [4–88], and FLUX [89].
112 This proliferation of architectures is precisely what calls for an evaluation methodology, such as ours,
113 that applies uniformly across model families using a single reference distribution and concept basis.

114 2.3 Datasets for Generative Image Models

115 The recent success of generative vision models is attributable in large part to the confluence of
116 abundant computation and large-scale internet datasets [90–92]. LAION-5B [93] occupies a central
117 place in this landscape, underpinning the training of open-source text-to-image models including
118 SD and its derivatives. Scraped from Common Crawl [94], it comprises over 5B image-caption
119 pairs, 2.3B of them in English; other prominent corpora include COYO-700M [95] and Conceptual
120 Captions [96], at 700M and 3M pairs, respectively.

121 The prominence of LAION-5B has been accompanied by mounting concern over its biases [97–101].
122 Filtering efforts notwithstanding, harmful content persisted [97–99]—including NSFW material [100]
123 and hundreds of CSAM instances [101]—prompting the dataset’s temporary withdrawal from official
124 channels. The corpus is further affected by low-quality images [102] and by internet-style captions
125 (e.g., product descriptions) that diverge from how users actually prompt trained models [103]. These
126 same distributional irregularities—noisy, underspecified, or only weakly aligned captions among
127 them—bear directly on our analysis: they are a factor our framework can connect to concept-level

generation failures, and, equally, a source of natural-versus-generated discrepancy that any such analysis must be careful to interpret.

2.4 Concept Discovery and Sparse Coding in Generative Image Models

Dictionary learning seeks sparse representations of input data in which each sample is reconstructed as a linear combination of a small number of dictionary atoms [104–106]. Grounded in compressed-sensing theory [107–108], the field has evolved from early vector-quantization methods [109] into a rich array of formulations, including non-negative matrix factorization [110–111], sparse PCA [112], and K-SVD [113]. More recent developments encompass online methods [114], structured sparsity [115], and theoretical recovery guarantees [116–117], alongside increasingly tight connections to deep learning [118–119].

Sparse coding has likewise been enlisted to study the emergence of high-level concepts within diffusion models [120]. Earlier generative architectures were made interpretable through concept-bottleneck models, which require human intervention at training time [121], and through post-hoc detectors that retrofit concept supervision [122]. Both families, however, presuppose a fixed set of human-defined concepts and therefore cannot register trends the user has not anticipated in advance—a constraint our unsupervised approach removes by construction, at the cost of a concept vocabulary that is learned rather than curated.

Aggregate metrics—precision, recall, density, and coverage [35–123]—together with latent density scores that predict sample quality from a model’s latent space [124], have also been proposed for evaluating generative image models. Effective as they are at exposing distributional gaps in aggregate, they remain silent about *which* specific concepts are under- or over-represented.

Collectively, these gaps motivate the scalable, unsupervised framework we develop for systematically identifying and quantifying concept-level failure modes in generative image models [125].

3 Formalizing Conceptual Blindspots in Generative Models

We begin by formalizing the notion of *conceptual blindspots*: systematic discrepancies between the conceptual content of natural images and that of a model’s generated outputs. The formalization serves two ends. First, it makes explicit the idealized conditions under which per-concept comparison would exactly recover a model’s conceptual bias, and in doing so yields a quantitative measure—the energy difference—that characterizes which concepts a generative model under- or over-represents relative to a reference image distribution (illustrated in Fig. 2). Second, it supplies the vocabulary in which our operational estimator is later stated. We stress at the outset that the derivation is a motivating idealization rather than an exact account of the empirical pipeline: it invokes standard assumptions from the identifiability literature [126–130], notably an invertible data-generating process, that the DINOv2+SAE instantiation only approximates. Reassuringly, we find that meaningful phenomenology is elicited even when these assumptions are not met exactly, which speaks to the practical robustness of the resulting diagnostic.

Definition 1 (Data-Generating Process). Let $\mathcal{C} \subset \mathbb{R}^K$ denote a latent space with a Boltzmann prior $p(\mathbf{c}) = \exp(-E(\mathbf{c}))Z^{-1}$, where $E(\cdot)$ denotes an energy function that linearly decomposes over individual latents and Z is the corresponding partition function, i.e., $E(\mathbf{c}) = \sum_k E(c_k)$ and hence $p(\mathbf{c}) = \prod_k p_k(c_k)$. A data-generating process (DGP) is an invertible function $G : \mathcal{C} \rightarrow \mathcal{X}$ that maps the latents $\mathbf{c} \in \mathcal{C}$ to observations $\mathbf{x} \in \mathcal{X}$, i.e., $\mathbf{x} = G(\mathbf{c})$.

For notational economy, we let $p(\cdot)$ denote both the latent density $p(\mathbf{c})$ and its pushforward to image space $p(\mathbf{x})$, where $\mathbf{x} = G(\mathbf{c})$. This overloading is well defined precisely because the invertibility of G induces, via the change-of-variables formula, a valid distribution over $\mathcal{X}$. Intuitively, the individual dimensions of the latent space correspond to the **concepts** underlying the data distribution $P_{\mathcal{X}}$, itself defined over an observation space of images $\mathcal{X}$. Distinct latents may thus encode concepts such as color, shape, size, or location [131–132]. The data-generating process is further taken to associate a text description $\mathbf{t} \in \mathcal{T}$ with any image sampled from the data distribution, an associ-

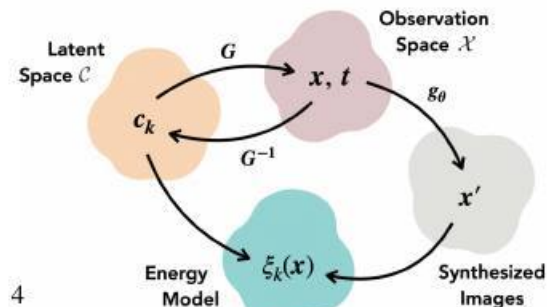


Figure 2: **Schematic of our framework.** Latent concepts $c_k \in \mathcal{C}$ are mapped to observations $(\mathbf{x}, \mathbf{t})$ through the (idealized, invertible) DGP; the gen-

181 ation we assume to exist but do not model ex-
 182 plicitly. These text descriptions constitute the
 183 training signal for a text-conditioned **generative**
 184 **image model** g_θ with parameters θ : trained
 185 on image-text pairs, g_θ maps a noise signal
 186 $\eta \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ together with a scene description
 187 t to an image realizing that description.

188 Under this formulation, defining conceptual
 189 blindspots in g_θ reduces to contrasting the prob-
 190 ability mass a concept receives under the data-
 191 generating process with the mass it receives un-
 192 der the model. To make the contrast operational,
 193 we take an evaluation set of natural image-text pairs $(\mathcal{D}_\mathcal{X}, \mathcal{D}_\mathcal{T})$ and construct a corresponding set
 194 of synthetic images $\mathcal{D}'_\mathcal{X}$ by conditioning g_θ on the same text descriptions. To estimate a concept's
 195 probability of occurrence, we introduce an **energy model** $\xi : \mathcal{X} \rightarrow \mathbb{R}^d$ mapping an image to a
 196 d -dimensional vector whose k^{th} coordinate is a scalar energy for the k^{th} concept, i.e., $\xi_k(\mathbf{x}) = E(c_k)$.
 197 Exact recovery of these energies is licensed by the invertibility assumption on the data-generating
 198 process; in practice we approximate them, as detailed in Sec. 4. The probability the data-generating
 199 process assigns to an image $\mathbf{x}$ is then $p_k(\mathbf{x}) \propto \exp(-\sum_k \xi_k(\mathbf{x}))$ (with Z_ξ the partition function),
 200 which aggregates to the population-level estimate $p_k(\mathcal{D}_\mathcal{X}) = \prod_{\mathbf{x} \in \mathcal{D}_\mathcal{X}} p_k(\mathbf{x})$. Composing this esti-
 201 mate with the sigmoid map $\sigma(\cdot)$, we define below the average energy difference for the k^{th} concept
 202 between the datasets $\mathcal{D}_\mathcal{X}$ and $\mathcal{D}'_\mathcal{X}$.

203 **Definition 2 (Energy Difference).** Let $\mathbf{x} \in \mathcal{D}_\mathcal{X}$ denote a real image sampled from the data-
 204 generating process G , and let $\mathbf{x}' \in \mathcal{D}'_\mathcal{X}$ be a synthetic image generated by the model g_θ . Let
 205 $\xi_k : \mathcal{X} \rightarrow \mathbb{R}$ denote the energy assigned to the k^{th} concept by the energy model ξ . We define the
 206 energy difference for concept k as:

$$\begin{aligned} \delta_{g_\theta \leftrightarrow G}(k) &= \sigma(\mathbb{E}_{\mathbf{x}'}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x}}[\xi_k(\mathbf{x})]) \\ &= \frac{p_k(\mathcal{D}'_\mathcal{X})}{p_k(\mathcal{D}_\mathcal{X}) + p_k(\mathcal{D}'_\mathcal{X})}, \end{aligned} \quad (1)$$

207 where the expectations are taken over $\mathcal{D}'_\mathcal{X}$ and $\mathcal{D}_\mathcal{X}$, respectively, and $p_k(\mathcal{D}) \propto \exp(-\sum_{\mathbf{x} \in \mathcal{D}} \xi_k(\mathbf{x}))$
 208 denotes the unnormalized conceptual probability mass of dataset $\mathcal{D}$ under concept k .

209 The energy difference for the k^{th} concept thus expresses, as a bounded quantity in $(0, 1)$, the relative
 210 probability that the concept occurs in a set of observations (here, $\mathcal{D}'_\mathcal{X}$) as against a baseline dataset
 211 (here, $\mathcal{D}_\mathcal{X}$). This measure furnishes the operational basis for our definition of conceptual blindspots.

212 **Definition 3 (Suppressed / Exaggerated Conceptual Blindspots).** Given a generative image model
 213 g_θ , we say, compared to the data-generating process G , c_k is a suppressed conceptual blindspot in
 214 the model if $\delta_{g_\theta \leftrightarrow G}(k) < \lambda_{\min}$ and exaggerated if $\delta_{g_\theta \leftrightarrow G}(k) > \lambda_{\max}$.

215 In sum, a conceptual blindspot is a concept whose likelihood of occurrence in generated images
 216 departs markedly—through either suppression or exaggeration—from its prevalence in the reference
 217 distribution. Suppressed concepts exhibit disproportionately low activation ($\delta(k) < \lambda_{\min}$); exagger-
 218 ated concepts are systematically over-represented ($\delta(k) > \lambda_{\max}$). Throughout our analysis we fix the
 219 thresholds $\lambda_{\min} = 0.1$ and $\lambda_{\max} = 0.9$ to demarcate the two regimes.

220 This definition may be read as a fine-grained refinement of “mode collapse” as studied in prior work
 221 (e.g., Bau et al. [39]), the distinction lying in the granularity of the analysis. Whereas mode collapse
 222 concerns the suppressed or exaggerated odds of generating *entire images*, our formulation isolates
 223 shifts in the odds of *specific concepts*. Concretely, when a model fails to produce images of an object
 224 set against a `white background`, we classify that concept as a suppressed conceptual blindspot.

225 4 Method: Operationalizing the Definition of Conceptual Blindspots

226 We now operationalize the formalization of Sec. 3 as a concrete pipeline for identifying conceptual
 227 blindspots in a generative model g_θ . The formalization calls for two constituent objects: (i) a set
 228 of images sampled from g_θ that admit direct comparison with images drawn from the reference

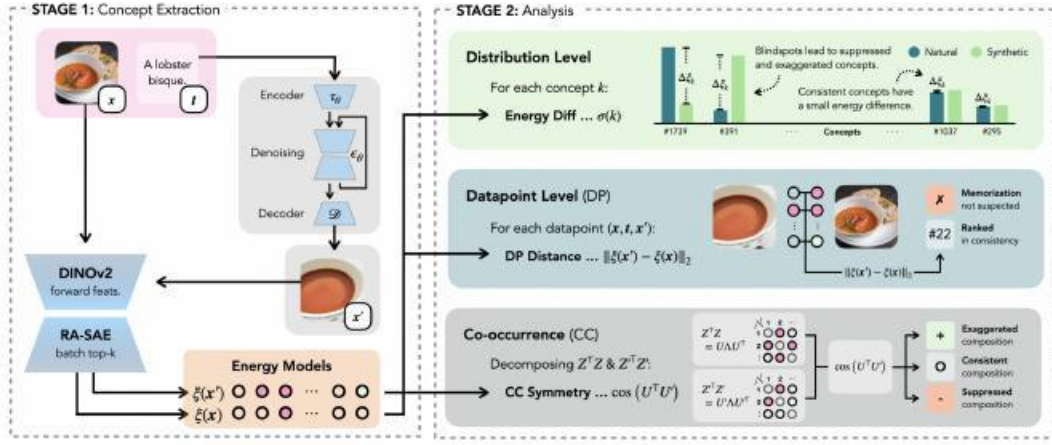


Figure 3: **Concept extraction pipeline.** For a triplet (x, t, x') , concepts are extracted in two steps: each image’s DINOv2 features are computed and then mapped through the RA-SAE into a sparse concept embedding, yielding the energy representations $\xi(x)$ and $\xi(x')$. The k^{th} coordinate $\xi_k(x) = E(c_k)$ records the energy of the k^{th} concept, and it is the per-concept difference between distribution, and (ii) an energy model that renders the comparison quantitative. The remainder of this section specifies each in turn, together with the sparse-coding step that supplies the interpretable concept basis. For notation, $\|\cdot\|_F$ denotes the Frobenius norm and $\|\cdot\|_0$ the number of non-zero entries (the ℓ_0 pseudo-norm). For a vector or matrix $\mathbf{X}$, the relation $\mathbf{X} \geq 0$ denotes element-wise non-negativity. For $n > 0$, we write $[n] := \{1, \dots, n\}$ and denote the i -th row of a matrix $\mathbf{A}$ by $\mathbf{A}_i$.

From Prompts to Latent Representations. Identifying conceptual blindspots in a model g_θ proceeds by comparing a dataset $\mathcal{D}_\mathcal{X}$ of image-caption pairs (x, t) drawn from the reference distribution against their synthetic counterparts, which g_θ produces from the same captions, so that any concept-level difference is attributable to the generator rather than to a change of prompt.

Concretely, for each caption t we synthesize a counterpart image x' with a pretrained text-to-image generator $g_\theta : \mathcal{T} \rightarrow \mathcal{X}$, realized as a denoising diffusion probabilistic model (DDPM) [78, 86, 87]. Generation unfolds in latent space along a reverse diffusion trajectory $(\gamma_t)_{t=0}^T$:

$$\gamma_T \sim \mathcal{N}(0, I), \quad \gamma_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(\gamma_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \varepsilon_\theta(\gamma_t, t, c) \right) + \sigma_t \eta_t, \quad \text{and} \quad \eta_t \sim \mathcal{N}(0, I),$$

where $\alpha_t \in (0, 1)$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$ follow the standard cosine noise schedule. The terminal latent γ_0 is decoded by a pretrained VAE to yield the synthetic image $x' = \text{VAE}(\gamma_0)$. For the remainder of the paper we treat g_θ as a black-box map from prompts to images, $t \mapsto x'$, as our diagnostic depends only on its input–output behavior; this is what renders the method agnostic to the internals of the generator under study.

Defining the Energy Model. Our choice of energy model rests on a body of work establishing that self-supervised learning can, under suitable conditions, invert the data-generating process and recover its underlying energy function up to a linear transformation [126, 128, 133, 134]; accordingly, we adopt DINOv2 [135], a state-of-the-art self-supervised model, as our feature extractor. Because the number of concepts underlying the DGP is expected to exceed the dimensionality of the feature space [51, 136], we train sparse autoencoders (SAEs) on these features to recover the subspaces corresponding to individual concepts [50, 56, 137–139]. The rationale is direct: if DINOv2 encodes the concepts underlying the generative process along approximately orthogonal directions—consistent with the independence assumption of Def. 1—then an SAE trained on its features can disentangle those concepts onto individual latent dimensions [136]. The activation the SAE assigns to a dimension then serves as our approximation of the ground-truth energy for the concept that dimension encodes. The diagnostic is thus defined relative to the concept vocabulary that DINOv2 and the SAE jointly expose: concepts these components represent poorly fall outside its reach, a dependence we return to in Sec. 6.

Formally, let $f : \mathcal{X} \rightarrow \mathbb{R}^d$ denote the feature-extraction module (the DINOv2 model), from which we extract features $\mathbf{a} = f(x) \in \mathbb{R}^d$ for both natural and synthetic images drawn from the datasets $\mathcal{D}_\mathcal{X}, \mathcal{D}'_\mathcal{X}$. Writing n for the number of image-text pairs, we stack the real and generated features into matrices $\mathbf{A}, \mathbf{A}' \in \mathbb{R}^{n \times d}$. Each feature vector is then decomposed into a sparse combination

of learned concept atoms by the SAE. Specifically, let $D \in \mathbb{R}^{d \times K'}$ be a dictionary of K' concept vectors and $\Psi : \mathbb{R}^d \rightarrow \mathbb{R}^{K'}$ the SAE encoder mapping input features to sparse codes. Applying Ψ row-wise produces the activation matrix $Z = \Psi(A) \in \mathbb{R}^{n \times K'}$, whose i -th row $z_i = \Psi(a_i)$ is the concept decomposition of the corresponding image. The decoder reconstructs features as $ZD^\top$, and the SAE is trained to minimize reconstruction error subject to joint sparsity and non-negativity constraints:

$$\min_{\Psi, D} \|A - \Psi(A)D^\top\|_F^2 \quad \text{s.t.} \quad \Psi(A) \geq 0, \|\Psi(A)_i\|_0 \ll K' \quad \forall i \in [n]. \quad (2)$$

Vanilla SAEs are prone to drifting toward arbitrarily oriented dictionaries, which renders downstream analyses acutely sensitive to the random seed—a liability for a diagnostic intended to be compared across models. To remove this instability, and thereby make our study reproducible and seed-independent, we employ the *archetypal* SAE (RA-SAE) [56] under a TOP-K sparsity constraint [137]. RA-SAE constrains each dictionary atom to a convex combination of the training data. Concretely, we factorize $D = WA$ with $W \in \Omega_{K',n}$, the set of row-stochastic matrices in $\mathbb{R}^{K' \times n}$:

$$\Omega_{K',n} := \{W \in \mathbb{R}^{K' \times n} \mid W \geq 0, W\mathbf{1} = \mathbf{1}\}. \quad (3)$$

Consequently, every atom D_i lies in the convex hull of the data, $\text{conv}(A)$, and any reconstruction $ZD^\top$ resides within the conic hull $\text{cone}(A)$. This construction keeps the learned concepts faithful to the support of the data distribution [56]. Once trained, the SAE yields a matched pair of sparse codes: Z for real images and Z' for their generated counterparts. *Because the two codes are expressed over a single shared, interpretable concept basis, they render prompt-conditioned visual semantics directly comparable, and their per-concept activations supply the energies on which the blindspot analysis operates.* The method therefore constitutes a structured pipeline that, given a prompt and its associated real image (t, x) , returns two sparse concept vectors (z, z') and so enables a direct, basis-aligned comparison of real and generated visual content.

In summary: starting from a captioned image (t, x) , we synthesize a counterpart x' and map both images—through the vision encoder followed by the trained SAE—into a shared sparse concept space, obtaining interpretable representations (z, z') that form the basis for evaluating the conceptual shifts the generative process induces.

5 Results

We analyze four generative image models trained on LAION-5B—SD 1.5, SD 2.1, PixArt, and Kandinsky—over $|\mathcal{X}| = 10,000$ image-text pairs and their corresponding generations (Fig. 4). The analysis proceeds at three complementary levels of granularity (Fig. 3): a • **distribution-level** evaluation reveals suppressed and exaggerated concepts; a • **datapoint-level** analysis surfaces failures attributable to caption ambiguity, omission, and memorization; and a • **compositional** analysis exposes subtle distortions in the geometry of concept co-occurrence.

At the center of the analysis sits an interactive exploratory tool (Fig. 5): a web-based interface organized around a UMAP projection of concept representations that supports the visualization and comparison of concept-level energy differences. The tool is publicly available at a software repository, together with pre-computed energy-difference data for all four evaluated models, and each analysis reported below is drawn from the insights it enables. Its principal functionalities are threefold:

- **Contrasting models and architectures.** For each evaluated model, the tool renders a UMAP visualization spanning all 32,000 RA-SAE concepts, each scatter point denoting an individual concept color-coded by its energy difference.



Figure 4: **Representative image-caption pairs** from LAION-5B, each shown alongside the images generated from the same caption by SD 1.5, SD 2.1, PixArt, and Kandinsky.

- **Inspecting concepts.** Every concept carries a detail card reporting its key statistics, representative real and generated images (x , x'), and visualized co-occurrence patterns.
- **Exploring blindspots.** Beyond the UMAP and per-concept views, the tool surfaces global rankings of suppressed and exaggerated concepts, foregrounding the most pronounced blindspots.

Rather than exhaustively pursuing any single phenomenon, we report a set of high-level findings that together demonstrate the breadth of the tool and invite the more targeted, customizable exploration it is designed to support.

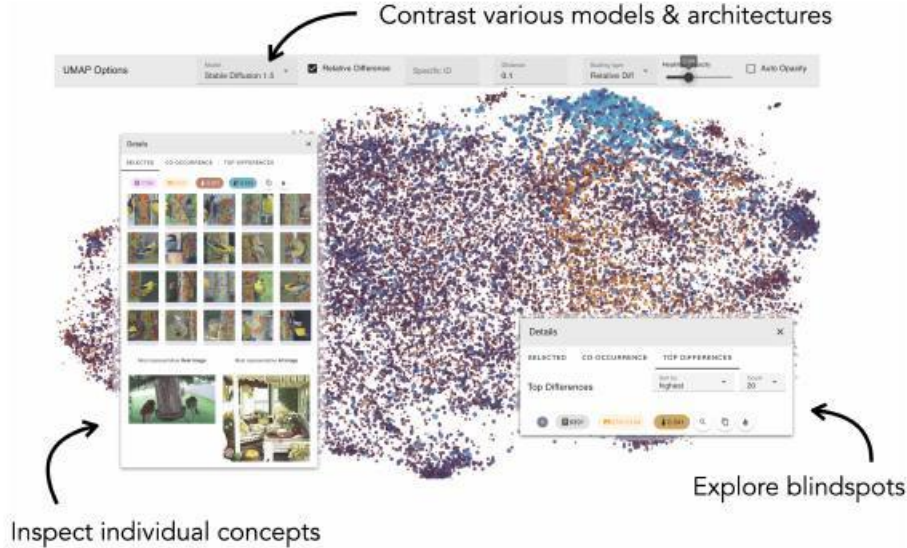


Figure 5: **Overview of the exploratory tool.** For each evaluated model, the web interface displays a UMAP projection in which every dot denotes a concept color-coded by its energy difference. Selecting a concept opens a detail panel presenting summary statistics, the most representative natural and generated images x and x' , and an ordered list of the concept’s co-occurrences, alongside global rankings of the most pronounced blindspots.

5.1 Concept Coverage Is Systematically Skewed Toward Suppression

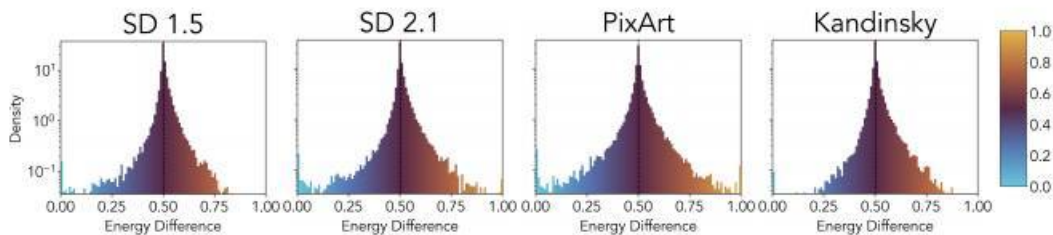


Figure 6: **Concept energy distribution.** Log-scale histograms of the energy differences $\delta(k)$ across all 32,000 concepts, contrasting the natural and synthesized distributions for each evaluated model. Mass toward the left corresponds to suppressed (under-represented) concepts, mass toward the right to exaggerated (over-represented) concepts. Every model exhibits a heavy-tailed, left-skewed profile, indicating a consistent bias toward suppression.

To characterize the disparities between the generative models and the natural image distribution, we first analyze the marginal energy difference $\delta(k)$ across the 32,000 concepts learned by the RA-SAE. As defined in Sec. 4, this quantity captures the relative prevalence of each concept in the synthesized versus the natural image set: $\delta(k) < 0.1$ marks concept k as under-represented (suppressed) in the generated images, whereas $\delta(k) > 0.9$ marks it as over-represented (exaggerated). Fig. 6 plots the distribution of $\delta(k)$ for each of the four models. Every model exhibits a heavy-tailed histogram with substantial mass at both extremes—evidence that discrepancies in concept coverage are systematic

rather than incidental. The asymmetry is the salient feature: across all four models the left tail, corresponding to suppressed concepts, is consistently denser and longer than the right, indicating a pervasive bias toward suppression over exaggeration. The negative skewness of all four distributions makes this quantitative: Skewness = -0.54 for SD 2.1, -0.40 for both SD 1.5 and PixArt, and -0.23 for Kandinsky.

While every model displays both suppressed and exaggerated concepts, their detailed profiles differ in informative ways. PixArt, for instance, exhibits a markedly wider spread, consistent with a more heavily suppressed concept distribution. The uniform left-skew nonetheless establishes a tendency toward concept omission shared across all four models; its finer structure is the subject of the sections that follow.

5.2 Structure and Specificity of Conceptual Blindspots

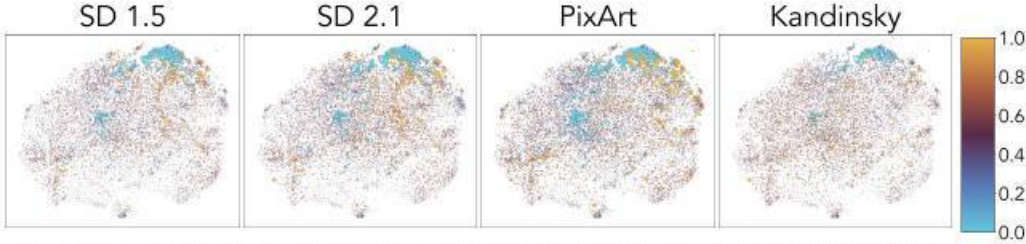


Figure 7: **Structure of concept energy differences.** UMAP visualizations of all 32,000 concepts, colored by their energy difference $\delta(k)$ between the natural and synthesized distributions (suppressed concepts at the blue end of the scale, exaggerated ones at the yellow end). The emergent clusters indicate that conceptual blindspots are spatially structured rather than randomly scattered across concept space.

Whereas the previous section quantified marginal discrepancies in concept frequency, we now interrogate their global structure, embedding the full set of 32,000 concepts into two dimensions via UMAP applied to the sparse codes and coloring each concept by its $\delta(\cdot)$ value. As Fig. 7 shows, distinct concept clusters emerge for every model. These clusters frequently coincide with contiguous regions of conceptual blindspots—most saliently among suppressed (blue) concepts—indicating that blindspots are spatially structured rather than randomly distributed, a pattern consistent with shared biases rooted in either the training distribution or architectural priors. To gauge how consistently these patterns recur across models, we examine both the magnitude and the structure of the concept-level $\delta(\cdot)$ values. Fig. 8 reports scatter plots and pairwise Pearson correlation coefficients between the $\delta(k)$ vector of SD 1.5 and those of the remaining models. SD 1.5 and SD 2.1 correlate strongly ($r = 0.82$), as their shared architecture and training pipeline would lead one to expect. Their correlations with PixArt and Kandinsky are substantially weaker— $r = 0.41$ and $r = 0.46$, respectively—indicating that these architectures emphasize markedly different regions of concept space.

Taken together, these results indicate that conceptual blindspots fall into two regimes: some are shared across models—plausibly a consequence of properties common to the training data—while

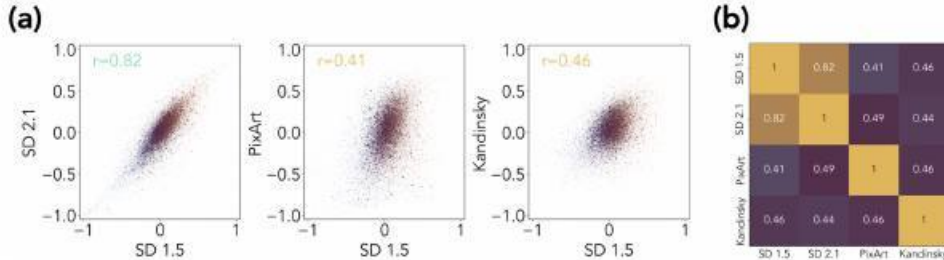


Figure 8: **Cross-model concept energy correlation.** Pairwise scatter plots of $\delta(k)$ across all four evaluated models, with Pearson correlation coefficients reported at the top left. The strong alignment between SD 1.5 and SD 2.1 contrasts with the weaker correlations among the other architectures, consistent with the presence of model-specific blindspots alongside shared ones. **Correlation matrix of conceptual blindspots.** Heatmap of the pairwise Pearson correlation coefficients of δ across all models, summarizing the degree to which blindspots are shared.

others are strongly model-specific, arising from idiosyncrasies of training dynamics or model capacity. This dichotomy motivates the sections that follow, in which we examine both the blindspots common across models and those particular to individual models.

5.3 Qualitative Examples of Blindspots

We now examine concrete examples of both suppressed and exaggerated blindspots to characterize which concepts populate each regime. Fig. 9a presents a blindspot suppressed across all four models: none reproduces the concept **solid white on documents**. Although the caption refers to the concept explicitly, none of the generated images realizes the intended visual semantics, indicating that this region of concept space is systematically under-sampled across the entire model set. Fig. 9b, by contrast, isolates a model-specific blindspot: the concept **pan** is faithfully rendered by three of the four models yet conspicuously absent from Kandinsky’s generations. This juxtaposition concretely illustrates the two regimes identified quantitatively in Sec. 5.2 where cross-model agreement proved high for some concepts and limited for others.

Conversely, Fig. 10 presents a case of exaggeration, in which the concept **shadow under animal** is disproportionately emphasized in the generated images. Although any single such shadow is plausible, their uniformly pronounced rendering across models—set against the far more nuanced and variable occurrences in natural images—is symptomatic of an overactive prior. Notably, a deliberate search for concepts exaggerated by a single model in isolation yielded no clear instances, suggesting that, in contrast to suppression, exaggeration tends to be shared across the evaluated models rather than model-specific.

In sum, these examples make concrete how conceptual blindspots manifest in generated outputs and show that our energy-based diagnostic surfaces both shared and model-specific failure modes. Notably, it exposes non-obvious limitations—most strikingly, the consistent failure to reproduce clear or solid background elements such as whitespaces on documents across every model. That this failure recurs across architectures points to a cause in shared training-data biases or common architectural constraints rather than in the idiosyncrasies of any single model.

Informative as these aggregate analyses are, they raise a sharper question: do blindspots arise only in aggregate across many samples, or do they persist at the level of individual datapoints? Answering it lets

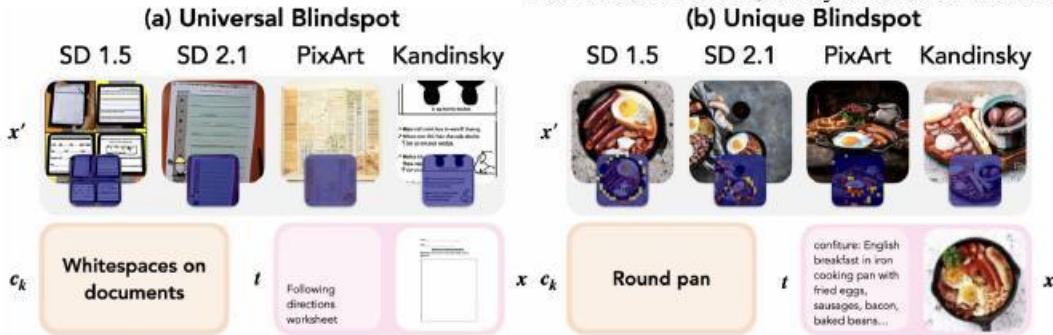


Figure 9: ● **Examples of suppressed conceptual blindspots.** The natural image x , with its representative concept c_k and caption t , shown alongside four synthesized images x' . (a) A blindspot shared across all evaluated models; (b) a blindspot present only in Kandinsky.



Figure 10: ● **Example of an exaggerated conceptual blindspot.** Four synthesized images $D_{x'}$, archetypal for the concept shadow under animal, shown with the corresponding natural image D_x and the caption t used to generate them. Note the accentuated, nearly solid black shadows

us interrogate the mechanisms behind blindspots more directly, and to begin disentangling prompt misinterpretation from latent memorization.

5.4 Investigating Datapoint-level Energy Difference: From Incongruent to Memorized Images

To move beyond population-level statistics, we inspect the individual natural-versus-generated image pairs whose $\delta(\cdot)$ values, averaged across all concepts, are largest and smallest. This analysis renders individual successes and failures directly legible, and it reveals that the failures frequently trace to one of two mechanisms: caption ambiguity or memorization artifacts. Fig. 11a shows pairs whose average $\delta(\cdot)$ is near zero, so that the generated image is conceptually near-indistinguishable from its original. Qualitative inspection indicates that this alignment arises not from faithful capture of the caption’s semantics but from replication of memorized templates: the recurrence of near-identical visual structures (e.g., outlines of clothing or object arrangements) is consistent with the model reproducing over-frequent patterns from the training data. Fig. 11b, conversely, shows pairs among the largest $\delta(\cdot)$ values, which signal pronounced conceptual divergence between the synthesized and natural images. Some such discrepancies are attributable to underspecified or noisy captions—an important confound, since a generation may follow its caption faithfully while still differing from the original image; others, however, expose genuine blindspots, in which the caption describes a clear concept that is present in $\mathcal{D}_{\mathcal{X}}$ yet the model fails to realize it in $\mathcal{D}'_{\mathcal{X}}$. This latter case indicates that, even when language grounding appears adequate, certain concepts lie beyond the model’s generative reach.



Figure 11: • **Datapoint-level conceptual alignment.** (a) Pairs with minimal energy difference, where the model appears to reproduce memorized training patterns. (b) Pairs with large energy difference, where pronounced concept divergence arises from caption ambiguity or from a genuine limitation of the model.

5.5 Analyzing Post-Training Effects

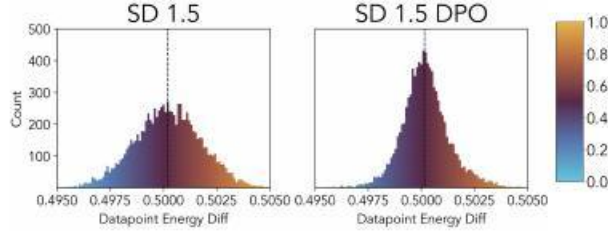


Figure 12: **Effect of DPO on concept fidelity.** Histogram of the datapoint-wise energy differences between the synthesized and natural distributions for SD 1.5, with and without DPO. The DPO variant shifts mass toward smaller differences, attaining both a lower median and a narrower spread.

Post-training protocols such as safety fine-tuning have been argued to reduce the diversity of model generations [140]. Because our pipeline isolates the discrepancies between a model’s generations and the reference distribution, it provides a natural instrument for quantifying the effect of DPO, a widely adopted safety fine-tuning protocol [141]. We compare two variants of SD 1.5—one trained with DPO and one without. For each image pair $(\mathcal{D}_X, \mathcal{D}'_X)$, we compute the ℓ_2 norm of the difference between their concept energy vectors, $\|\xi(\mathcal{D}'_X) - \xi(\mathcal{D}_X)\|_2$. Fig. 12 presents a histogram of these datapoint-wise energy differences. The DPO-trained model attains both a lower median and a narrower spread, indicating that its generated concepts align more consistently with the reference distribution. DPO thus appears to regularize the model’s concept distribution, steering outputs toward closer agreement with the semantic content of the observed inputs. While our analysis does not disentangle the specific inductive biases DPO introduces, these results furnish direct empirical evidence that its human-preference objective indirectly improves the match to the reference distribution. Tellingly, the effect attenuates *both over- and under-activation* of individual concepts relative to the natural baseline, rather than merely shifting the distribution in one direction. More broadly, the experiment illustrates the utility of our pipeline for characterizing the downstream effects of post-training interventions—not only on output quality, but on how such interventions reshape the conceptual geometry of a model’s output space.

5.6 Conceptual Misalignment as a Function of Empirical Frequency

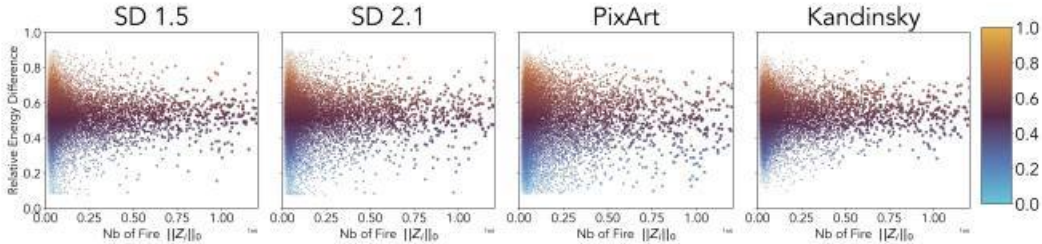


Figure 13: **Concept fidelity across the frequency spectrum.** Scatter plots relating concept frequency (x-axis) to energy difference (y-axis) for all four evaluated models. Each point denotes a concept, with marker size proportional to its activation frequency in the natural data. High-frequency concepts cluster at low energy difference; the largest discrepancies concentrate among rare concepts.

We hypothesized above that conceptual blindspots are not solely architectural artifacts, but may also emerge as a consequence of distributional peculiarities of individual features. We now test this hypothesis empirically, asking whether concepts rarely activated in natural images—those in the long tail of the data distribution—are disproportionately prone to becoming blindspots. Concretely, we process the natural dataset $\mathcal{D}_X$ through the trained SAE and compute, for each concept k , its empirical frequency $\|\mathbf{Z}_{:,k}\|_0$, where $\mathbf{Z}_{:,k}$ denotes the activations of concept k across all images, and relate this frequency to the absolute energy difference observed across generated outputs. Fig. 13 visualizes the relationship for all evaluated models. A clear and consistent pattern emerges: high-frequency concepts exhibit low energy discrepancies, whereas rare concepts—suppressed ones in particular ($\delta(k) < 0.5$)—incur substantial alignment errors. This indicates that many blindspots are tied to

456 systematic long-tail effects in the concept distribution rather than to randomness or isolated model
457 quirks, and it suggests that remedying them may call not merely for architectural changes but for
458 targeted interventions such as data reweighting or augmentation.

459 6 Discussion

460 Our analysis uncovers multiple conceptual blindspots across four widely used generative image
461 models and, taken together with our targeted stress tests, indicates that these blindspots are systematic
462 rather than incidental. The findings reported here represent an initial mapping of the phenomenon:
463 each individual result could sustain a dedicated investigation of its own. Rather than exhausting any
464 single question, we have deliberately foregrounded the versatility of the method and its exploratory
465 tool, which together support—out of the box—the systematic identification of concepts a model
466 struggles to generate, the detection of memorization artifacts, the discovery of datapoints with
467 insufficient captions, the quantification of post-training effects, and the characterization of conceptual
468 shifts across architectures. This breadth leaves ample room for follow-up work to deepen the analysis,
469 broaden the set of evaluated architectures, and probe the root causes of conceptual blindspots. A
470 particularly promising direction is the development of hierarchical concept representations within
471 RA-SAE, which would enable a more nuanced, compositional analysis of the relational failures that
472 single-concept scores cannot fully resolve.

473 **Limitations.** Several limitations bound the scope of our claims. First, because concept extraction
474 and representation rest on DINOv2 and the RA-SAE, the analysis is confined to the concepts these
475 components encode; concepts they represent poorly—or that are relational or compositional rather
476 than visually localized—will elude detection, and we accordingly interpret single-concept scores with
477 more confidence than the higher-order co-occurrence structure examined in the appendix. Second,
478 because the reference distribution comprises LAION-derived image–caption pairs whose captions
479 may be noisy, underspecified, or weakly aligned with the image, a measured energy difference
480 can reflect caption–image mismatch rather than a failure of the model to follow the prompt; our
481 datapoint-level analysis (Sec. 5.4) separates these cases only qualitatively, and our targeted stress tests
482 corroborate a representative subset rather than the full ranking. Third, we generate a single image
483 per caption, so a portion of each concept’s energy difference is attributable to stochastic sampling
484 rather than to a stable model property; our concentration bound constrains estimation error at the
485 population level but not this per-datapoint variance. Finally, although a sample of 10,000 images is
486 substantial, it may not fully resolve the long tail of rare concepts, their co-occurrences, or higher-order
487 compositional statistics. These considerations delimit, rather than undercut, the framework’s central
488 use: a scalable, reproducible means of surfacing candidate concept-level failures for closer study.

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Appendix

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A Experimental Setup

This section details the experimental setup underlying our analysis of four widely used generative image models—SD 1.5/2.1, Kandinsky, and PixArt—all trained on LAION-5B or its subsets and derivatives. The code is available at a software repository.

A.1 Observation Space

The observation space, which constitutes our reference distribution of natural images, is constructed by sampling 10,000 image-text pairs from the LAION-5B dataset [93]. Because of concerns over CSAM and other unsafe content, the original release is no longer available; we therefore draw from a substitute release of a filtered subset with additional removal of unsafe content, available at <https://huggingface.co/datasets/laion/relaion2B-en-research-safe>

The sampling procedure comprises three stages: (1) loading the full LAION dataset via the Hugging Face datasets library; (2) validating each candidate—confirming proper URL structure and resource availability through HTTP HEAD requests; and (3) drawing random samples with replacement until the target of 10,000 valid samples is reached. The procedure yields D_G as a set of (x, t) tuples and ensures that the observation space consists exclusively of accessible image-text pairs—a nontrivial requirement, given that a large fraction of the original image URLs has become unavailable since release. Additional examples of synthesized images are provided in Appendix H

A.2 Synthesized Images

For each of the four evaluated models, we generate a synthetic dataset D_{g_θ} in one-to-one correspondence with D_G , thereby forming triplets (x, x', t) . Concretely, for each of the 10,000 image-text pairs (x, t) in D_G , we condition the generator g_θ on the caption t to synthesize a counterpart image x' . Conditioning every model on the same captions is what makes the resulting energy differences directly comparable across architectures.

The synthesis follows the standard text-to-image generation pipeline of each architecture, implemented with the Hugging Face diffusers library and loaded at mixed precision (fp16). All synthetic images are generated at 512×512 pixel resolution using each model’s default parameters.

Stable Diffusion 1.5. We use the checkpoint from <https://huggingface.co/benjamin-paine/stable-diffusion-v1-5> a mirror of the deprecated <https://huggingface.co/runwayml/stable-diffusion-v1-5>. Inference employs 50 denoising steps at a fixed guidance scale of 7.5.

Stable Diffusion 1.5 + DPO. The DPO variant of SD 1.5, used in the analysis of Sec. 5.5 mirrors the baseline SD 1.5 implementation but substitutes a DPO-trained UNet from <https://huggingface.co/mhdang/dpo-sd1.5-text2image-v1>

Stable Diffusion 2.1. We use the checkpoint from <https://huggingface.co/stabilityai/stable-diffusion-2-1>. Inference employs 50 denoising steps at a fixed guidance scale of 7.5.

Kandinsky. We use the checkpoint from <https://huggingface.co/kandinsky-community/kandinsky-2-1>. Inference employs 100 denoising steps at a fixed guidance scale of 4.0.

PixArt. We use the checkpoint from <https://huggingface.co/PixArt-alpha/PixArt-XL-2-1024-MS>. Inference employs 50 denoising steps at a fixed guidance scale of 7.5.

A.3 Distribution Level Analysis

Section 5.1 For each evaluated model, we compute the energy differences $\delta(\cdot)$ across all 32,000 concepts, applying a sigmoid transformation with temperature $T = 0.8$ during normalization. The resulting values are visualized as log-scale density histograms with 100 bins spanning $[0, 1]$.

Section 5.2 We embed the complete set of 32,000 concepts into two dimensions via UMAP applied to the sparse concept codes, each point representing an individual concept colored by its energy

difference $\delta(\cdot)$ so as to foreground both suppressed and exaggerated blindspots. To quantify cross-model consistency, we compute pairwise Pearson correlation coefficients between the $\delta(\cdot)$ vectors of all model pairs, yielding both scatter plots and correlation matrices. This analysis establishes whether blindspots cluster in concept space and separates model-specific from shared patterns of conceptual blindspots.

Section 5.3 We rank all 32,000 concepts by their energy difference $\delta(\cdot)$ and manually inspect the extrema—the suppressed and exaggerated blindspots. For suppressed blindspots we select concepts with $\delta(\cdot) < 0.1$; for exaggerated blindspots, those with $\delta(\cdot) < 0.9$, consistent with the thresholds that demarcate the two blindspot regimes. Each presented example is manually annotated with a textual description of the underlying concept, obtained by inspecting its most activating images and their spatial attention patterns. Ongoing efforts to automate this concept interpretation are outlined in Appendix D

Section 5.5 We compare SD 1.5 with and without DPO as follows: for each image pair (x, x') , we compute the ℓ_2 norm of the difference between their concept energy vectors, $\|\xi(x') - \xi(x)\|_2$, applying a sigmoid transformation with temperature $T = 0.8$ to the element-wise differences before averaging. This yields datapoint-wise energy differences that quantify how far each generated image deviates from its natural counterpart in concept space, which we then visualize as overlapping histograms contrasting the two model variants.

Section 5.6 For each concept c_k , we count its empirical frequency $\|Z_{:,i}\|_0$ —the number of non-zero activations across the natural dataset—and apply a sigmoid normalization with temperature $T = 0.4$ to the energy differences $\delta(\cdot)$. The relationship is visualized as scatter plots whose x-axis is the empirical concept frequency and whose y-axis is the sigmoid-transformed energy difference; marker sizes are proportional to activation frequency and marker colors to the magnitude of the energy difference.

A.4 • Datapoint Level Analysis

Section 5.4 For each image pair (x, x') , we compute the ℓ_2 norm of the difference between their concept energy vectors, $\|\xi(x') - \xi(x)\|_2$, yielding a scalar measure of conceptual divergence per pair. Ranking the samples by this quantity places memorization artifacts at the minimal-divergence extreme and pronounced conceptual failures at the maximal-divergence extreme, so that specific failure modes can be inspected qualitatively in a targeted way.

A.5 • Co-occurrence Analysis

Appendix F For both the natural and synthesized datasets D_G and $D_{g\theta}$, we analyze concept co-occurrence through the co-activation matrix $Z^T Z$, which encodes the pairwise correlations in concept usage. We recover the dominant conceptual directions via an eigendecomposition of this matrix and assess the alignment between the natural and synthetic co-occurrence structures using cosine-similarity heatmaps between the top-100 eigenvectors of each. These 100×100 similarity matrices are rendered as square heatmaps in which perfect diagonal alignment would signify identical principal concept axes, whereas off-diagonal structure reveals rotations and mismatches in the compositional geometry.

1002 B Custom RA-SAE Training Details

1003 This section summarizes the training procedure for the custom archetypal SAE (RA-SAE) [56] that
1004 serves as the core instrument of our pipeline. The model is open-sourced at a software repository.

1005 **Dataset.** The autoencoder is trained on the complete ImageNet-1k training split (≈ 1.28 M RGB
1006 images). DINOv2 [135] encodes each image into 261 visual tokens, which are supplied to the SAE
1007 without class or position embeddings. The total number of training tokens is therefore $50 \times 1.28\text{M} \times$
1008 $261 \approx 1.67 \times 10^{10}$.

1009 **Dictionary.** The dictionary comprises 32,000 concept dimensions—to our knowledge the largest
1010 RA-SAE reported to date. Sparsity is enforced by top- k masking with $k = 5$, which zeroes all but the
1011 five largest activations per input. Weights are initialized with the Xavier/Glorot scheme, and training
1012 is conducted at mixed precision (fp16), with the final ten epochs performed at full precision.

1013 **Optimizer and schedule.** The model is trained for 50 epochs with the AdamW optimizer ($\beta_1 = 0.9$,
1014 $\beta_2 = 0.999$) and weight decay 10^{-5} . A linear warm-up over the first 5% of steps is followed by
1015 cosine decay from $\eta_{\max} = 5 \times 10^{-4}$ to $\eta_{\text{final}} = 10^{-6}$. The objective combines an MSE reconstruction
1016 loss with an auxiliary term, weighted by $\lambda = 10^{-5}$, that penalizes activations never entering the top- k
1017 set.

1018 C Computational Resources

1019 This section summarizes the GPU resources expended on the training and experiments supporting
1020 this paper—approximately 202 GPU-hours in total on NVIDIA H100 and H200 hardware.

1021 **RA-SAE.** Trained in approximately 24 GPU-hours on three NVIDIA H100s.

1022 **Synthesized Images.** Generating the full $D_{g\theta}$ (see Appendix A) required roughly 5 hours per
1023 generator when distributed across four NVIDIA H200 GPUs; across five generators, this amounted to
1024 approximately 100 GPU-hours on NVIDIA H200s.

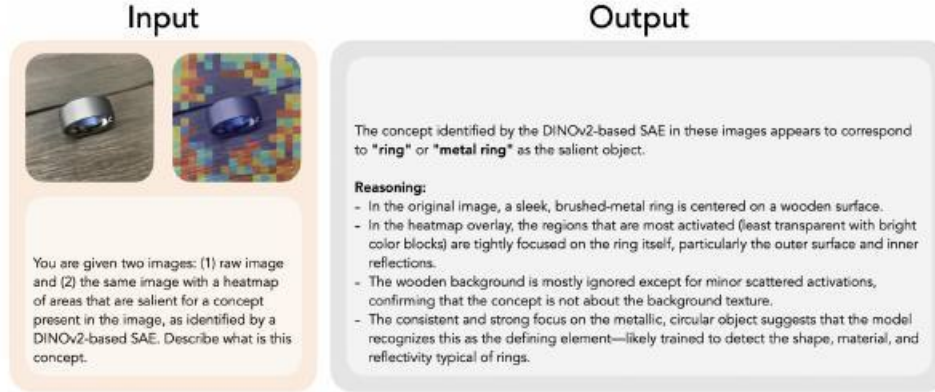
1025 **● Distribution-Level Analysis.** Extracting distribution-level energy differences required about 3
1026 hours per generator on a single NVIDIA H200 GPU (total ~ 15 GPU-hours).

1027 **● Datapoint-Level Analysis.** Computing datapoint-level energy differences, ranking concepts per
1028 datapoint, and ranking datapoints per concept likewise required approximately 3 hours per generator
1029 on one H200 GPU (total ~ 15 GPU-hours).

1030 D Autointerpretability

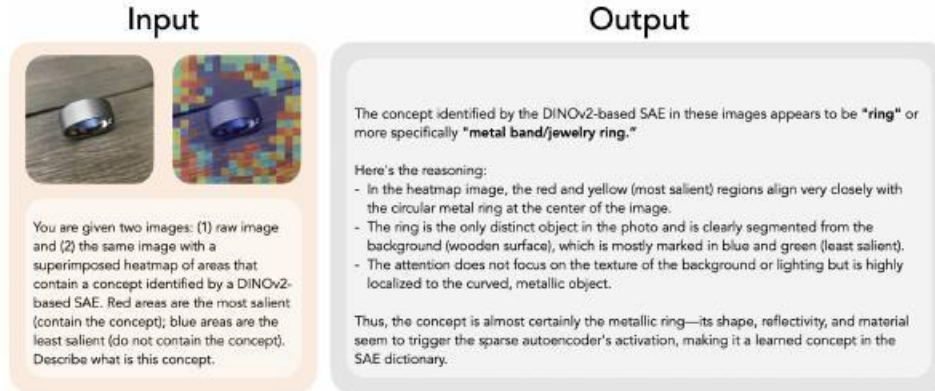
1031 To automate the interpretation of concepts identified by the RA-SAE, we conducted a qualitative
 1032 evaluation of the ability of vision-language models (VLMs) to describe them. Specifically, we queried
 1033 ChatGPT 4o via the OpenAI API with a range of prompts and image formats, each depicting an image
 1034 x'' in which the target concept is salient, and recorded the VLM’s description. This section reports
 1035 the findings of that exploration. For evaluation we permitted the VLM to emit its full reasoning
 1036 trace; a scalable deployment would suppress this trace through a straightforward prompt modification
 1037 requesting the description alone.

1038 We first supplied the VLM with two images: a raw version of x'' and a copy of x'' with a superimposed
 1039 heatmap localizing the intensity of the concept:



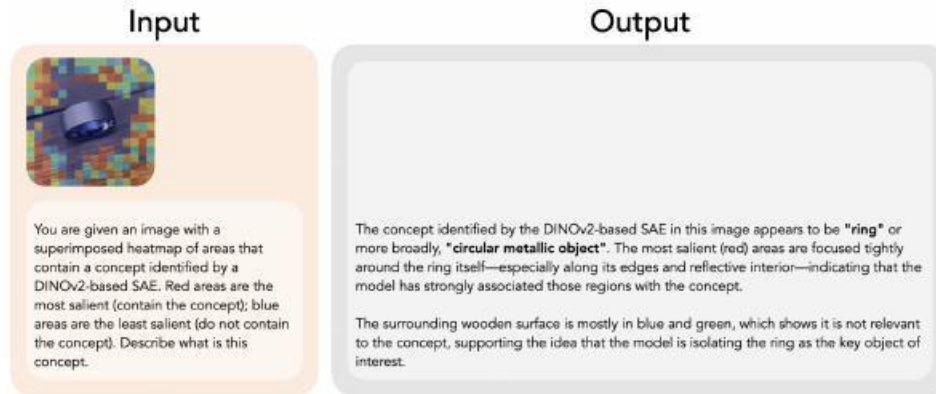
1040 As the example shows, the VLM failed to interpret the heatmap visualization correctly: rather than
 1041 describing the wood texture in the background—the genuinely salient region—it described the ring
 1042 in the foreground, to which the RA-SAE assigned no activation.

1043 To determine whether this reflected a prompting deficiency, we repeated the scenario with an expanded
 1044 prompt explaining the heatmap in greater detail:



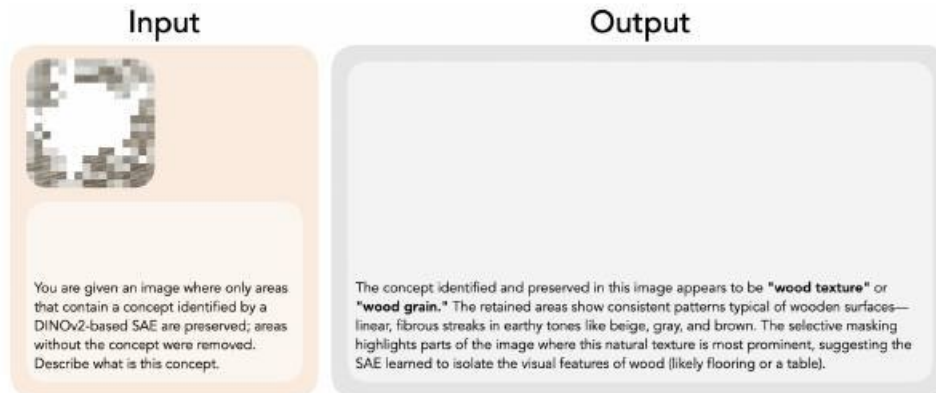
1045 This modification did not resolve the issue: the VLM continued to identify the ring as the prominent
 1046 concept. Several further prompting strategies—including ones describing the heatmap in still greater
 1047 depth, for instance by supplying the code used to generate it—yielded similar results.

1048 We next provided the VLM with only the heatmap-superimposed x'' , together with a comparable
 1049 prompt explaining how the heatmap should be interpreted:



1050 Once again the VLM failed to identify the wooden background as the prominent concept, attending
 1051 instead to the ring; as before, varying the prompting strategy for this image format left the results
 1052 largely unchanged.

1053 These observations lead us to conclude that the heatmap alone is insufficient to direct the VLM's
 1054 attention to the salient region, as the presence of less relevant regions distracts the model. To remedy
 1055 this, we replaced the heatmap-superimposed x'' with an alpha-masked version of x'' :



1056 As the example shows, the VLM now correctly identified wood texture as the dominant concept,
 1057 and we observed consistent success across a range of concepts and images.

1058 The image format of x'' and the prompt shown above yielded the most reliable results in our
 1059 qualitative evaluation. This evaluation is nonetheless bounded on two counts: it is qualitative, owing
 1060 to the absence of ground-truth concept annotations, and it is restricted to a single VLM. We anticipate
 1061 that future work on the autointerpretability of SAE concepts can build on and extend this analysis
 1062 toward a quantitative protocol.

E Additional Results: Model-Specific Blindspots

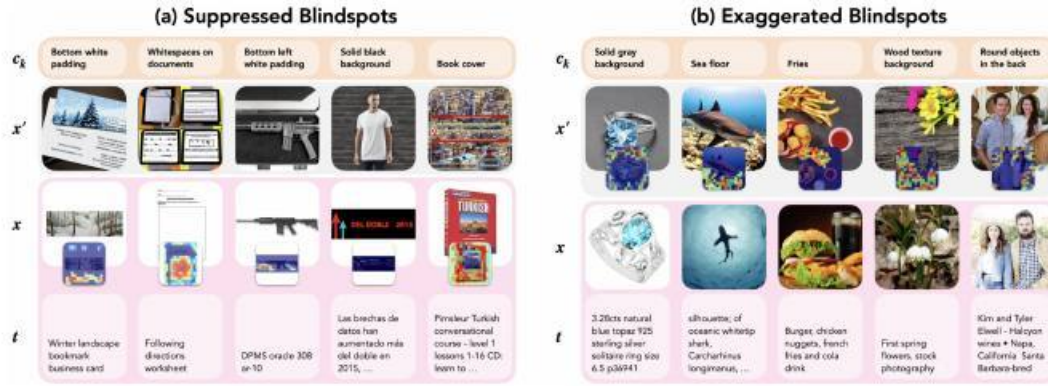


Figure 14: Examples of conceptual blindspots in **Stable Diffusion 1.5**. For each concept we display the prototypical image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by highest absolute activation, with the concept’s spatial heatmap superimposed.

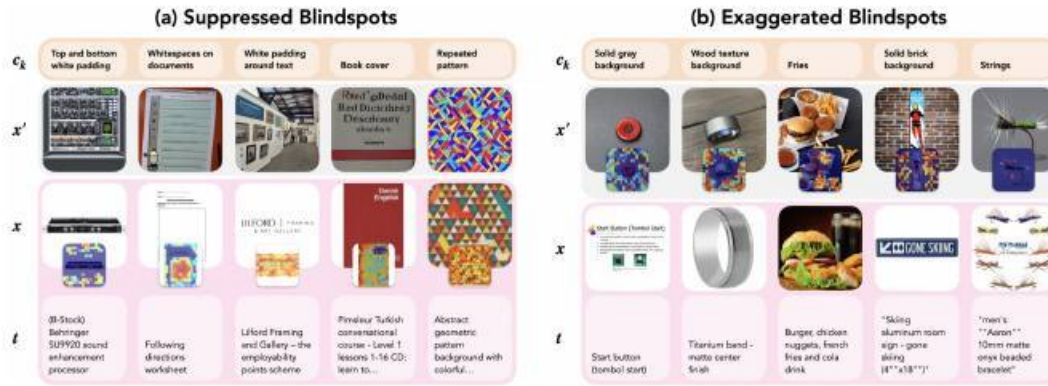


Figure 15: Examples of conceptual blindspots in **Stable Diffusion 2.1**. For each concept we display the prototypical image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by highest absolute activation, with the concept’s spatial heatmap superimposed.

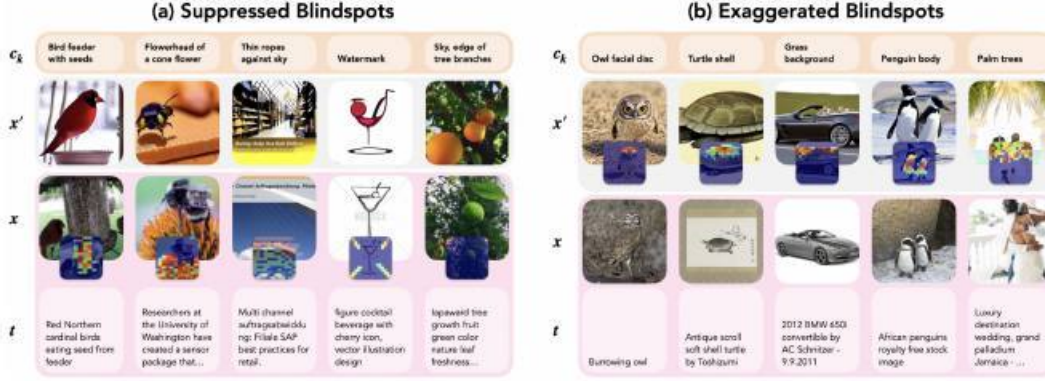


Figure 16: Examples of conceptual blindspots in **Kandinsky**. For each concept we display the prototypical image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by highest absolute activation, with the concept’s spatial heatmap superimposed.



Figure 17: Examples of conceptual blindspots in **PixArt**. For each concept we display the prototypical image—natural for suppressed blindspots, synthesized for exaggerated ones—selected by highest absolute activation, with the concept’s spatial heatmap superimposed.

F Additional Results: Higher-order Blindspots with Compositional Discrepancy

Our analysis has thus far centered on individual concept activations. Visual scenes, however, are rarely composed of isolated concepts; they are instead organized through rich co-occurrence patterns that encode compositional semantics. We now assess whether generative models capture this higher-order structure by analyzing the co-activation matrix $Z^T Z$, which encodes the pairwise correlations in concept usage. Because these are second-order statistics of the same concept scores analyzed above, the conclusions here are correspondingly more exploratory than the single-concept findings of the main text.

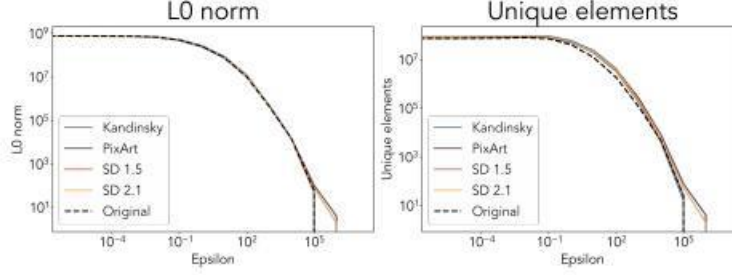


Figure 18: • **Sparsity and structural divergence.** Left: the ℓ_0 norm of the co-occurrence matrix ZZ^T as a function of the threshold ϵ , indicating how many entries remain active in each model. Right: the number of unique entries in the synthesized distribution relative to the natural distribution. Every evaluated model preserves the global sparsity structure yet diverges in the specific activation content.

Assessed at the level of binary structure, diffusion models approximate the global sparsity of the natural co-occurrence matrix with striking fidelity. As Fig. 18 (left) shows, the ℓ_0 norm of $Z^T Z$ —thresholded across a range of ϵ values—tracks closely between the natural and synthesized distributions for every model, indicating that the gross connectivity of the conceptual graph, i.e., which concepts co-activate at all, is well preserved. Formally, $Z^T Z$ may be regarded as the adjacency matrix of a weighted, undirected graph over concepts whose edge weights encode co-activation strength across the dataset.

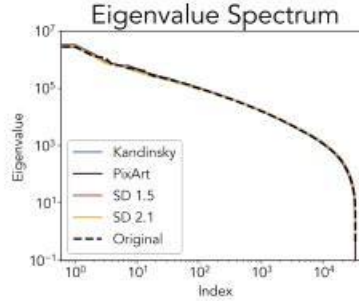


Figure 19: • **Spectral structure of co-occurrence.** Log-log plot of the eigenvalue spectra of the co-occurrence matrices ZZ^T for each model and for the natural distribution. Every evaluated model reproduces the heavy-tailed decay of the natural distribution.

The specific content of these co-activations, however, diverges markedly: as Fig. 18 (right) illustrates, a substantial fraction of the entries in the model-generated $Z^T Z$ are absent from the natural baseline. The capacity for compositionality is thus retained even as the identity of the active pairings shifts—plausibly reflecting model-specific inductive biases or training artifacts. To probe the internal structure of these patterns further, we turn to spectral analysis. Fig. 19 displays the eigenvalue spectra of the co-occurrence matrices for each model and for the natural distribution. All spectra exhibit heavy-tailed, approximately power-law decay, indicating that the generative models preserve the overall rank structure and the allocation of variance across conceptual dimensions.

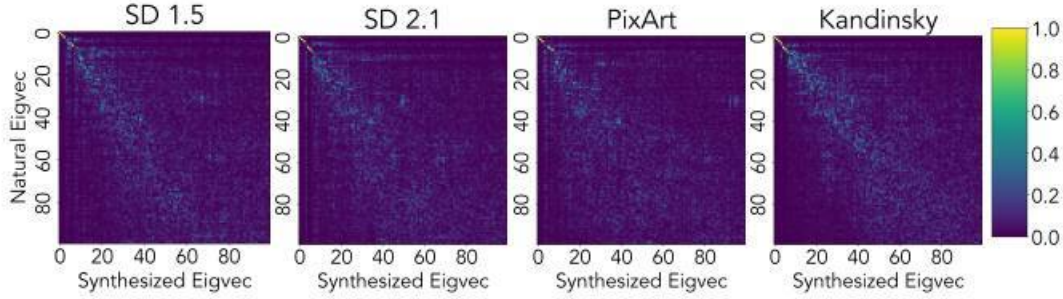


Figure 20: ● **Concept basis similarity.** Cosine-similarity heatmaps between the top 100 eigenvectors of the natural and synthesized co-occurrence matrices ZZ^T . The diagonal structure indicates alignment of the dominant conceptual directions, with the degree of alignment varying across the four models.

1089 We further examine the alignment of the dominant conceptual directions via cosine-similarity
 1090 heatmaps between the top 100 eigenvectors of the synthesized and natural co-occurrence matri-
 1091 ces (Fig. 20). While all models exhibit partial diagonal alignment—implying overlap in the principal
 1092 concept axes—the off-diagonal entries expose rotations and mismatches in the higher modes, consis-
 1093 tent with systematic deviations in compositional geometry.

1094 Together, these findings suggest that diffusion models approximate the global shape of concept
 1095 co-activation remarkably well yet deviate in subtle, structured ways when viewed through the spectral
 1096 lens. Such higher-order discrepancies may, in turn, underlie failures in generating coherent multi-
 1097 object scenes or relational concepts—a connection we regard as a hypothesis for future work rather
 1098 than one this analysis establishes.

1099 G Additional Results: Blindspot Stress Testing

1100 The blindspots surfaced by our method are candidate failures inferred from a single generation per
 1101 caption; to test whether they persist under deliberate, repeated prompting, we conducted a targeted
 1102 stress test. For each of a representative set of identified blindspots, we assembled a range of prompts
 1103 describing the concept, generated many images from them, and contrasted the outputs of a model for
 1104 which the concept was flagged as a blindspot against those of a model for which it was not.

1105 Concretely, ChatGPT-4o was prompted as follows: *I want to generate an image of the following*
 1106 *concept: "<blindspot>". Suggest 50 prompts highlighting this concept to be used as input for a*
 1107 *text-to-image model. Return these as a list of strings in Python.* For each prompt we generated five
 1108 images and analyzed them with our custom RA-SAE model (see Appendix B), ranking them by
 1109 the intensity with which the target concept appeared; every image was then manually reviewed to
 1110 determine whether the blindspot was successfully depicted.

1111 As the examples of suppressed concepts below show, fragments of the target concept occasionally
 1112 surfaced—a holder or string for the bird feeder blindspot, a round hole for the glossy DVD
 1113 disc blindspot—yet the flagged model consistently failed to render the full concept, even across
 1114 fifty distinct prompts and repeated sampling. That the failure survives this concerted effort, while
 1115 the comparison model renders the same concept readily, corroborates the method’s assessment on
 1116 this subset and provides direct, if targeted, evidence that the identified blindspots reflect genuine
 1117 generative limitations rather than artifacts of a single prompt or sample.

1118 G.1 Bird Feeder Blindspot in Kandinsky



Figure 21: Images generated from a variety of prompts involving the bird feeder concept during stress testing. Our method flagged this concept as a suppressed conceptual blindspot in Kandinsky, and the observed behavior bears this out: across the prompt set, the model fails to render a corresponding image. SD 1.5, for which the concept was not flagged, renders it successfully.



Figure 22: Images generated from a variety of prompts involving the `glossy DVD disc` concept during stress testing. Our method flagged this concept as a suppressed conceptual blindspot in SD 1.5, and the observed behavior bears this out: across the prompt set, the model fails to render a corresponding image. Kandinsky, for which the concept was not flagged, renders it successfully.

1120 H Additional Examples: Synthesized Images



Figure 23: Additional image-caption pairs from LAION-5B, each shown alongside the images generated from the same caption by SD 1.5, SD 2.1, PixArt, and Kandinsky.

1121 I Concentration bounds for δ

1122 In our experiments we estimate $\delta(k)$ from $n = 10,000$ paired samples per concept. Modest as this
 1123 budget is, whether it suffices for reliable estimation warrants a formal answer, which we supply
 1124 by deriving—via McDiarmid’s inequality [142]—a concentration bound on the empirical estimator
 1125 $\hat{\delta}_n(k)$. The bound establishes that accurate estimates of concept bias are attainable even from
 1126 comparatively few samples, addressing whether the population-level rankings we report are stable at
 1127 this sample size.

1128 **Theorem 4** (Concentration of $\hat{\delta}_n(k)$). *We assume that the concept score $\xi_k(\mathbf{x})$ takes values in $[a, b]$
 1129 almost surely for all images $\mathbf{x}$ drawn from either $\mathcal{D}_X$ or $\mathcal{D}'_X$. Let n paired samples $(\mathbf{x}_i, \mathbf{x}'_i)_{i=1}^n$ be
 1130 drawn independently with $\mathbf{x}_i \sim \mathcal{D}_X$ and $\mathbf{x}'_i \sim \mathcal{D}'_X$, and define the empirical estimator*

$$\hat{\delta}_n(k) := \sigma \left(\frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}_i) - \frac{1}{n} \sum_{i=1}^n \xi_k(\mathbf{x}'_i) \right).$$

1131 *Let $M := b - a$ and $L := M/4$. Then for every $\varepsilon > 0$, the deviation satisfies*

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right).$$

1132 *Proof.* The function $x \mapsto \sigma(x)$ is $1/4$ -Lipschitz, since $|\sigma'(x)| \leq 1/4$ for all x .

1133 Viewing $\hat{\delta}_n(k)$ as a function of the $2n$ independent variables $(\mathbf{x}_1, \dots, \mathbf{x}_n, \mathbf{x}'_1, \dots, \mathbf{x}'_n)$, changing a
 1134 single argument alters the inner difference of means by at most M/n , and the outer sigmoid scales this
 1135 by at most $1/4$. Hence, the bounded difference constant for each coordinate is $(M/n)(1/4) = L/n$.

1136 By McDiarmid’s inequality [142],

$$\mathbb{P} \left(\left| \hat{\delta}_n(k) - \delta(k) \right| > \varepsilon \right) \leq 2 \exp \left(-\frac{2\varepsilon^2}{\sum_{j=1}^{2n} (L/n)^2} \right) = 2 \exp \left(-\frac{2n\varepsilon^2}{L^2} \right),$$

1137 which proves the claim. □

1138 In practice the concept scores $\delta(k)$ are sparse: most concentrate near zero, and only a few reach
 1139 values as large as 10. The concentration bound guarantees that, even for the largest observed biases, a
 1140 sample size of $n = 10,000$ yields estimates $\hat{\delta}_n(k)$ deviating from the true value by no more than a
 1141 small ε with high probability. This justifies our sampling strategy at the population level, though we
 1142 note it bounds estimation error for the aggregated score and not the per-datapoint variance arising
 1143 from stochastic sampling of the generator.

1144 J Monotonicity and Calibration-Free Interpretation of δ

1145 Our aim in analyzing blindspots is to rank concepts by the severity of their generative bias, for which
 1146 we employ the score $\delta(k)$. It is natural to ask whether this score introduces distortions relative to more
 1147 direct quantities such as the energy gap or the odds ratio. The following result settles the question:
 1148 $\delta(k)$ is a strictly increasing reparameterization of both, and therefore inherits their ordering exactly,
 1149 so that no calibration is required when $\delta(k)$ is used to rank concepts.

1150 **Theorem 5** (Monotonicity and Calibration of $\delta_{g_\theta \leftrightarrow G}$). *For every concept index k define the energy*
 1151 *gap*

$$\Delta_k = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}'_X}[\xi_k(\mathbf{x}')] - \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_X}[\xi_k(\mathbf{x})],$$

1152 *the associated odds ratio $\rho_k = \exp(\Delta_k)$, and the energy-difference score*

$$\delta(k) = \frac{1}{1 + \exp(-\Delta_k)} = \frac{\rho_k}{1 + \rho_k}.$$

1153 *Then $\delta(k)$ is a strictly increasing bijection of both Δ_k and ρ_k , so ranking concepts by any one of*
 1154 *$\delta(k)$, Δ_k , or ρ_k produces exactly the same ordering.*

1155 *Proof.* The logistic sigmoid satisfies $\sigma'(x) = \sigma(x)(1 - \sigma(x)) > 0, \forall x \in \mathbb{R}$; hence σ and therefore
 1156 $\delta(k) = \sigma(\Delta_k)$ grow strictly with Δ_k . Because the exponential map is also strictly increasing and
 1157 bijective $\mathbb{R} \rightarrow (0, \infty)$, setting $\rho_k = \exp(\Delta_k)$ preserves order and gives $\Delta_k = \log \rho_k$. Substituting
 1158 this identity into σ yields $\delta(k) = \sigma(\log \rho_k) = \rho_k / (1 + \rho_k)$, which is the composition of two
 1159 strictly increasing bijections and is therefore itself strictly increasing and bijective in ρ_k . Since strict
 1160 monotonic functions never reverse inequalities, the three quantities share the same total order over
 1161 concepts.

1162

□

1163 Consequently, ranking concepts by $\delta(\cdot)$ is strictly equivalent to ranking them by the energy gap or
 1164 by the conceptual generation odds ρ_k : no calibration is necessary, as all three quantities induce the
 1165 same total ordering over concepts. The bounded score $\delta(\cdot)$ is thus a convenient reparameterization
 1166 for reporting, with no cost to the ordering it induces.

K Stability of FID Under SAE Embeddings

In this section we establish a quantitative relationship between the Fréchet Inception Distance (FID) computed in the original activation space (of dimension d) and the FID computed after applying a (potentially overcomplete) SAE dictionary $D \in \mathbb{R}^{k \times d}$ with $k \gg d$. The purpose is to certify that the SAE embedding on which our diagnostic operates does not materially distort the standard distributional metric. Throughout, we assume D has orthonormal columns but is not necessarily square, i.e.

$$D^\top D = I_d, \quad \text{while} \quad DD^\top \neq I_k.$$

We begin by recalling a simple fact: when D is orthogonal and non-overcomplete ($k = d$), it induces an isometry between A and Z , so that FID is preserved exactly. As this case is unrealistic for an SAE, we then turn to the overcomplete regime and show that FID can be bounded in terms of the extreme singular values of D . Throughout, we work with the Wasserstein-2 metric $\mathcal{W}_2$, noting that FID is precisely $\mathcal{W}_2^2$ specialized to Gaussians.

For a probability measure μ on $\mathbb{R}^d$ we write $D_\# \mu$ for its push-forward under D , i.e. $D_\# \mu(z) = \mu(D^{-1}z)$. Denote by $\sigma_{\min}$ and $\sigma_{\max}$ the minimal and maximal singular values of D , equivalently the square-roots of the extremal eigenvalues of $DD^\top$:

$$\sigma_{\min}^2 I_k \preceq DD^\top \preceq \sigma_{\max}^2 I_k.$$

Empirically one typically finds $\sigma_{\min}, \sigma_{\max} \approx 1$, though the proof relies on no such assumption. We begin with a simple lemma for the non-overcomplete case.

Lemma 1 (Isometry under exact orthogonality). *Suppose $k = d$ and $D^\top D = DD^\top = I_d$. Then D is an isometry: $\|Dv\|_2 = \|v\|_2$ for all $v \in \mathbb{R}^d$. Consequently, for any probability measures μ, ν on $\mathbb{R}^d$ with finite second moment,*

$$\mathcal{W}_2(D_\# \mu, D_\# \nu) = \mathcal{W}_2(\mu, \nu).$$

Proof. Orthogonality of D implies preservation of the Euclidean norm, and push-forward commutes with the map inside the $\mathcal{W}_2$ infimum; the integrand is unchanged, so the infimum value is identical. $\square$

This case is nonetheless unrealistic, since SAEs rely on overcompleteness to extract meaningful, interpretable concepts. In the overcomplete regime D is no longer orthonormal, yet it can still be column-orthonormal; we exploit this property to bound FID in terms of the extremal singular values of $D^\top D$.

Theorem 6 (FID under column orthogonal embeddings). *Let $D \in \mathbb{R}^{k \times d}$ satisfy $D^\top D = I_d$ and denote by $0 < \sigma_{\min} \leq \sigma_{\max}$ the extreme singular values of $DD^\top$. Given two data matrices $A, A' \in \mathbb{R}^{n \times d}$ (rows are sample vectors), set $Z = AD^\top \in \mathbb{R}^{n \times k}$ and $Z' = A'D^\top \in \mathbb{R}^{n \times k}$. Then*

$$\sigma_{\min}^2 \text{FID}(A, A') \leq \text{FID}(Z, Z') \leq \sigma_{\max}^2 \text{FID}(A, A').$$

Proof. Write μ for the empirical measure of A and ν for that of A' , i.e.

$$\mu = \frac{1}{n} \sum_{i=1}^n \delta_{A_{i,:}}, \quad \nu = \frac{1}{n} \sum_{i=1}^n \delta_{A'_{i,:}}.$$

For any coupling $\pi \in \Pi(\mu, \nu)$ (i.e. a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ, ν) we have, by the extremal singular value bound,

$$\sigma_{\min}^2 \|x - y\|_2^2 \leq \|D(x - y)\|_2^2 \leq \sigma_{\max}^2 \|x - y\|_2^2, \quad \forall (x, y) \in \mathbb{R}^d \times \mathbb{R}^d.$$

Integrating with respect to an arbitrary coupling $\pi \in \Pi(\mu, \nu)$ yields

$$\sigma_{\min}^2 \int \|x - y\|_2^2 d\pi \leq \int \|D(x - y)\|_2^2 d\pi \leq \sigma_{\max}^2 \int \|x - y\|_2^2 d\pi.$$

The middle integral is exactly the transport cost of the pushed-forward coupling $(D \times D)_\# \pi$ between $\mu_D := D_\# \mu$ and $\nu_D := D_\# \nu$. Because the inequalities hold for every π , they hold in particular

1203 for the optimal couplings attaining $\mathcal{W}_2(\boldsymbol{\mu}, \boldsymbol{\nu})$ and $\mathcal{W}_2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D)$, though these two optima need not
 1204 coincide. Taking the infimum over $\boldsymbol{\pi}$ term-wise makes this explicit:

$$\sigma_{\min}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi} \leq \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|D(\mathbf{x} - \mathbf{y})\|_2^2 d\boldsymbol{\pi} \leq \sigma_{\max}^2 \inf_{\boldsymbol{\pi} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \int \|\mathbf{x} - \mathbf{y}\|_2^2 d\boldsymbol{\pi}.$$

1205 Hence

$$\sigma_{\min}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}) \leq \mathcal{W}_2^2(\boldsymbol{\mu}_D, \boldsymbol{\nu}_D) \leq \sigma_{\max}^2 \mathcal{W}_2^2(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

1206 Recognising $\text{FID}(\cdot, \cdot) = \mathcal{W}_2^2(\cdot, \cdot)$ for the Gaussian surrogate and plugging in $(\mathbf{A}, \mathbf{A}')$ (resp. $(\mathbf{Z}, \mathbf{Z}')$)
 1207 finishes the proof. $\square$

1208 In essence, Theorem 6 establishes that applying a column-orthogonal overcomplete SAE dictionary
 1209 cannot distort the Fréchet Inception Distance by more than the square of its extremal singular values.
 1210 When $\mathbf{D}$ is nearly orthogonal—and empirically we consistently found $\sigma_{\min}, \sigma_{\max} \approx 1$ —the result
 1211 implies that FID measured in the SAE feature space is essentially indistinguishable from the canonical
 1212 FID, so the interpretability afforded by the SAE comes at negligible cost to this distributional metric.
 1213 $\square$

1214